Grade: Check Plus

STERN SCHOOL OF BUSINESS
NEW YORK UNIVERSITY

EXECUTIVE MBA
DC CLASS OF 2027

COR1.1205.85 Prof. Aaron Tenenbein Statistics and Data Analysis

ASSIGNMENT 1

Attached Problem.

Due: February 24, 2026 before 6 PM

The probability distribution of the amount drawn from an ATM at a convenience store is tabulated below. In this table X = the amount withdrawn by a randomly selected customer and f(x) = P[X = x]. A customer can only withdraw the amounts indicated in the first column of the table below.

x	f(x)
10	**0.10**
20	**0.15**
40	**0.30**
60	**0.15**
80	**0.25**
100	**0.05**

a. What is the probability that a randomly selected customer withdraws at least $60 from the ATM?

Answer:

✓ Qa

Probability of at least $60 are withdrawn by a randomly selected customer equals:
f(>=60) = 0.15 + 0.25 + 0.05 = 0.45, or 45%

b. Show by calculation that the expected value withdrawn from the ATM is $50 and that the standard deviation of the amount withdrawn from this ATM is $26.27.

Part 1: Show by calculation E[X]=$50
Answer:

x	f(x)=P(X =x)	x·f(x)	x^2·f(x)
10	0.10	10(0.10)=1	100(0.10)=10
20	0.15	20(0.15)=3	400(0.15)=60
40	0.30	40(0.30)=12	1600(0.30)=480
60	0.15	60(0.15)=9	3600(0.15)=540

80	0.25	80(0.25)=20	6400(0.25)=1600
100	0.05	100(0.05)=5	10000(0.05)=500
	1.00	50.00	3190

✓ Qb

E[X]=∑xf(x)
1 + 3 + 12 + 9 + 20 + 5 = 50

E[X]=50

Part 2: Show the standard deviation is $26.27
Answer:
$x^2 \cdot f(x)$
10+60+480+540+1600+500=3190
$E[x^2]=3190$

Variance = $3190 - 50^2 = 3190 - 2500 = 690$

SD = $\sqrt{690} = 26.27$

c. The manager of this convenience score tells you that an expected value of $50 is not correct because it is impossible to withdraw $50 from the ATM. How would you answer the manager?

✓ Qc

Answer:

There is a difference between possible outcomes and expected value. An expected value is determined by looking at a long-run average across many customers, not a single withdrawal amount by a single customer.

The expected value is calculated specifically by taking a weighted average of the possible outcomes (withdrawal amounts).

d. What is the probability that the amount withdrawn from the ATM will fall within one standard deviation of the mean?

Answer:

The standard deviation is $26.27. So one standard deviation would be +/- 26.27 of the mean (expected value)

✓ Qd

The mean is 50.

Therefore, 50 + 26.27 = 76.27 , 50 - 26.27 = 23.73

The possible values within this interval range: 40, 60

40 has a probability of 0.3, 60 has a probability of 0.15

Therefore, the probability that the amount withdrawn from the ATM will fall within one standard deviation of the mean is 0.45, or 45%

e. What is the probability that the amount withdrawn from the ATM will fall within two standard deviations of the mean?

Answer:✓Qe

The standard deviation is $26.27. So two standard deviations would be +/- 52.54 of the mean (expected value)

The mean is 50.

Therefore, 50 + 52.54 = 102.54 , 50 - 52.54 = -2.54

The possible values within this interval range: 10, 20, 40, 60, 80, 100

The probabilities of each value: 0.1+0.15+0.3+0.15+0.25+0.05=1.00

Therefore, the probability that the amount withdrawn from the ATM will fall within two standard deviations of the mean is 1.00, or 100%